

TILAK PARAJULI

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AI/ML engineer and researcher working on evaluation reliability: I red-team frontier LLM agents, measure where automated judges fail, and build the evaluation gates that decide what reaches a human. I also own production for a multi-service agent platform on AWS.

PUBLICATIONS & RESEARCH

- Neither Blinding Nor a Jury, Neither Capability Nor Strategy: What Brings a Verifier to Reliability** 2026
J. Kozman, A. Waddepalli, T. Parajuli. Accepted, *NeurIPS 2026 Workshop on Trust-AI-Eval (TAE)*, 230 of 509 submissions. Extended version under review, *ICLR 2027*.
– 64,800 controlled verifications, four models, three domains. With the error held fixed, a model approves its own wrong answers **11.9 pp** more often than other models do; telling it who wrote the answer moves accuracy under 1 pp. Only execution grounding restores reliability (93.3% vs. 76.7% for the best ungrounded jury). Owned the mathematics domain (verifiers catch just 26.4% of wrong answers); built the prompt-parity test that keeps strategy a clean independent variable.
- Which Verifier Catches What? Mapping Error Severity to Verifier Strategy in Multi-Agent LLM Pipelines** 2026
T. Parajuli, J. Kozman, A. Waddepalli. Selected as a top AI/ML project, *SISTER 2026 summer research institute*. Code and traces public.
– Controlled fault injection into a plan/solve/verify pipeline: six paired conditions, 200 GSM-Hard problems, Claude Sonnet 4.6 and Qwen 2.5 72B. Removing the verifier raises execution-stage propagation to **27.3%** (Holm-adjusted $p < 0.0001$); CoT and rubric verifiers repair 95–98% of caught faults vs. 78.5% direct.

EXPERIENCE

- Data Scientist / AI Engineer** Jan 2026 – Present
Climate Clean Solutions & LowPropTax California, USA / Kathmandu, Nepal (Remote)
– Own production for a **7-service, 4-Lambda multi-agent platform** for property-tax appeals on AWS (ECS Fargate, Lambda, API Gateway v2, SQS/SNS, S3, CloudWatch) across dev and prod; **40 Terraform resource types**, GitHub Actions for build, test, deploy and release.
– Built the release path: dev pipelines record a signed image manifest per commit; prod promotes the exact digests dev ran, re-renders task definitions from declared config, waits for rollouts, and posts a per-service summary. **Zero manual AWS steps per release**.
– Traced a 100-property production upload across two codebases from CloudWatch and the database: proved which of 145 outcomes were correct, filed five portal defects with file and line, shipped four fixes the same day.
– Designed the agents' error contract (stable reason_code, client-safe reason, internal detail) across five services. **Least-privilege IAM**: every new AWS API simulated against the deploy role before a branch exists. Diagnosed OIDC and ECS task-protection incidents.
– Took over the income-approach valuation engine after a 13-defect rewrite; CI baseline **987 tests, 0 failures**. Built the per-appeal LLM cost model and provider routing used for pricing decisions.
- Red-Team & Evaluation Engineer** Sep 2026 – Present
Independent Contractor, AI Safety Evaluation Remote
– Design adversarial and edge-case scenarios for frontier LLM agents with **machine-verifiable pass/fail criteria** covering over-refusal, over-disclosure and **prompt injection**; run them across frontier models and report attack-success and solvability rates.
- AI/ML Research Fellow** Jun 2025 – Sep 2025
Fusemachines Nepal Kathmandu, Nepal
– Built an automated model-evaluation pipeline (MAE, RMSE, MAPE, SMAPE) deployed across forecasting models after peer review.
- RPA Developer** Feb 2025 – May 2025
Quickfox Consulting Kathmandu, Nepal
– Built an LLM-powered data-cleaning pipeline for legacy medical databases; built RPA bots for data entry and web scraping.
- AI Research Intern** Jun 2024 – Sep 2024
NAAMII, under Dr. Taman Upadhaya Kathmandu, Nepal
– Clinical and radiomics analysis for radiation oncology: permutation tests, ROC AUC, biomarker models.

PROJECTS

- Zeroth: Evaluation-Gated Clinical Multi-Agent Platform** | Python, FastAPI, Next.js, PostgreSQL, Pydantic 2026
– Genomics, imaging, radiomics and clinical-note agents produce findings; an evaluation gate returns PASS/FLAG/BLOCK and refuses to display any claim without provenance. **Security by design**: closed a login enumeration oracle, timing-safe scrypt auth, revocable sessions, per-source throttling, PHI allowlist, append-only audit log.

SKILLS

AI Safety & Security: red-teaming frontier agents, prompt injection, over-refusal/over-disclosure, LLM-as-judge evaluation, adversarial scenario design, threat modeling, least-privilege IAM, secrets management, auth hardening, PHI handling
Cloud & Infrastructure: AWS (ECS Fargate, Lambda, API Gateway, SQS/SNS, S3, ECR, CloudWatch, IAM, Secrets Manager, SSM, PrivateLink, Route 53), Terraform, Docker, MongoDB Atlas, DuckDB, Parquet
CI/CD & Release: GitHub Actions (OIDC to AWS, matrix builds, per-service test baselines, release manifests, image promotion by digest), permission simulation before deploy
LLM Engineering: RAG (Pinecone, Qdrant), fine-tuning (LoRA/PEFT), multi-agent systems, DeepInfra, Anthropic API, LangGraph, per-request cost accounting, PyTorch
Backend & Data: Python, FastAPI, Pydantic, pytest, boto3, SQS consumers with idempotency, structured logging, PostgreSQL, Redis, SQL, Bash, TypeScript

EDUCATION

- Bhaktapur Multiple Campus, Tribhuvan University** Dec 2019 – Oct 2024
B.Sc. in Computer Science and Information Technology, 78/100, First Division with Distinction Bhaktapur, Nepal
– Full Scholarship, Annual Nepal AI School (ANAIS 2025) · AI Fellow, Fusemachines AI Fellowship (2025), full scholarship